Final Report – Group 12

Public charging infrastructure and electric passenger cars in Dutch municipalities, 2022–2025

Project Group

Members:

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AI Statement

We acknowledge the use of Claude, ChatGPT, Gemini and Microsoft Copilot in preparing this submission. We used them for the following purposes: refining language (grammar, clarity), assisting with programming and debugging, and brainstorming ideas. We confirm that we have critically reviewed and verified all AI-assisted content, that the submitted work reflects our own understanding, and that we take full responsibility for its accuracy and originality.

Eline: I used the AI tool ChatGPT, to improve spelling, grammar and language in parts of the report. I also used ChatGPT for support with the coding process and to understand how GitHub works. All suggestions were reviewed and evaluated before being used in the final work.

Hendrik: I used Claude for processing data, assisting with programming and refining language.

Anne Jente: I used ChatGPT to help brainstorm ideas when choosing the project subject, and later in the process to refine grammar.

Matthijs: I used AI tools, mainly ChatGPT, to support the coding process and to help write, revise and improve parts of the report. I also used ChatGPT for guidance on GitHub and the pull request process and how this works.

Donna: I used AI tools (such as ChatGPT and Gemini) to assist in writing and improving parts of the text and to provide support during the coding process.

Introduction and motivation

Electric passenger cars are becoming more common in the Netherlands. The number of fully electric passenger cars owned by private persons and businesses grew from about 137,000 on 1 January 2022 to around 361,000 on 1 January 2025 (Centraal Bureau voor de Statistiek, 2023, 2026). All of these cars need a place to charge. Many owners can charge their cars at home, for example in their own driveway. However, in towns and cities many residents have no private parking space and depend on public charging points in the street (Wolterman et al., 2025).

Dutch policy follows the *Ladder van Laden* (charging ladder). This means that electric vehicle owners should charge on their own property where possible (Rijn, 2022). Only people without their own driveway or parking space can request a public charging point from their municipality. Municipalities are responsible for installing public charging points. Regional partnerships also support them, for example by organising the concessions for charging-point operators (Heijnen, 2022). The Nationale Agenda Laadinfrastructuur, which is part of the Dutch Klimaatakkoord, estimates that 1.7 million charging points will be needed in 2030 to charge all electric passenger cars. About 400,000 of these will need to be located in public spaces (Heijnen, 2022). To follow the development, the national government uses indicators such as the number of electric cars per charging point (Heijnen, 2022). Since April 2024, European regulations also link public charging infrastructure to the number of electric cars. For every registered battery-electric car, an EU member state must provide at least 1.3 kW of publicly accessible charging power (European Parliament & Council of the European Union, 2023).

Although these targets are at a national level, the actual installation of charging points is organized by municipalities. There are 342 municipalities in the Netherlands, each with its own policy and approach. The growth of public charging infrastructure can therefore differ between municipalities. When there are not enough charging points compared to the number of electric vehicles, residents without a private parking space may have trouble finding a place to charge their car. This could make the switch to electric driving less attractive to people. On the other hand, when charging points are installed faster than the demand increases, public money and space may be used where they are less needed.

The aim of this project is therefore to compare the development of electric passenger cars with the development of public charging points in every Dutch municipality between 2022 and 2025. For this comparison, the project uses the same indicator as the national government, the number of electric cars per public charging point. The analysis also takes differences between municipalities, such as size and population density, into account. The results show which municipalities have managed to keep their public charging infrastructure

in line with the growth of electric cars and which municipalities have not managed to keep up.

Research Objective

The objective of this project is to assess the extent to which the growth of public charging infrastructure in Dutch municipalities corresponds with the growth of electric passenger cars. Municipal-level electric vehicle (EV) registration data from 2022 to 2025 is combined with data on public charging-point availability over the same period. The analysis examines whether municipalities with larger or faster-growing EV fleets also have higher levels of public charging infrastructure and infrastructure growth. In this way, the project aims to identify the relationship between private EV adoption and public charging infrastructure, and to assess whether infrastructure provision is aligned with potential charging demand based on the number of electric passenger cars.

Research question

Main research question

To what extent has public charging infrastructure in Dutch municipalities kept pace with the growth of electric passenger cars between 2022 and 2025?

Sub-questions

1. How has the number of electric passenger cars developed across Dutch municipalities between 2022 and 2025?
2. How has the number of public charging points developed across Dutch municipalities between 2022 and 2025?
3. Which Dutch municipalities show a match or mismatch between charging infrastructure availability and EV adoption between the end of 2022 and the end of 2024?

Public charging infrastructure has *kept pace* when the number of electric cars per public charging point did not clearly rise. Because charging points can follow electric cars with a delay, a small rise is not counted as a mismatch; the introduction to sub-question 3 explains what counts as a match or a mismatch.

Sub-questions 1 and 2 each describe one dataset over its full period (2022–2025). Sub-question 3 and the main question compare both datasets at the same moments, which the data allow from the end of 2022 to the end of 2024 (see Data preparation).

Data Used

To answer the main research question and its subquestions, multiple sources were used. Four annual datasets from Centraal Bureau voor de Statistiek tracked the absolute number of electric passenger cars for Dutch municipalities over the years 2022-2025. These numbers include the cars of private persons and of businesses linked to one person, such as sole proprietorships (CBS: *natuurlijke personen en gekoppelde rechtspersonen*). Other company cars and lease cars are not included, because they are often registered at the address of the company instead of where the driver lives.

The number of public charging points per municipality was taken from the dataset *Aantallen laadpunten per gemeente* from the Rijksdienst voor Ondernemend Nederland (RVO). For each municipality and each year from 2022 to 2025, it gives the number of regular public charging points, regular semi-public charging points and fast charging points (> 50 kW).

The number of inhabitants per municipality on 1 January 2025 comes from CBS (table 03759ned) and is used to express electric cars and charging points per 1,000 inhabitants. The population density per municipality in 2025 (inhabitants per km² of land), also from CBS, is used to compare urban and rural municipalities.

The municipal borders of 2025 were obtained from CBS through PDOK (*CBS gebiedsindelingen*, generalised municipal borders). They are used to map the results per municipality.

Data Pipeline

The pipeline runs from the raw files in `data/` to the figures in `figures/` and the tables in this report. The analysis is in two Python modules in `src/` (`plot_maps.py` and `charging_vs_evs.py`), which the code cells in this report call. Two small scripts download the population and the municipal borders.

Step	What happens	Code
1. Collect	Read the CBS files with electric cars (1 January 2022–2025) and the RVO file with public charging points (December 2022–2025). Download the population (CBS OData API) and the municipal borders (PDOK WFS) once.	`src/download_population.py` , `src/download_municipal_borders.py`
2. Clean	Remove line breaks from column names and spaces used as thousands separators; give municipality names one spelling; add up municipalities that merged after 2022; sum the three types of public charging points.	`read_ev_file()` , `to_number()` , `name_key()` , `load_evs()` , `load_charge_points()` in `plot_maps.py`
3. Merge	Combine all sources into one table with one row per municipality on the 2025 borders (342 municipalities), checking that every municipality matches exactly once.	`load()` in `plot_maps.py`

Step	What happens	Code
4. Indicators	Calculate electric cars and charging points per 1,000 inhabitants, growth percentages and the number of electric cars per charging point. For sub-question 3, match the dates of both datasets and add population density.	`load()` in `plot_maps.py` and in `charging_vs_evs.py`
5. Analyse	Sub-questions 1 and 2: maps of levels and growth. Sub-question 3: change in EVs per charging point (with a 10% margin for delay), a linear regression of new charging points on new electric cars (both per 1,000 inhabitants), a comparison of four population-density groups and a check whether charging points only follow electric cars with a delay of one year.	`plot_ev_maps()`, `plot_charge_point_maps()`, `evs_per_charge_point()`, `match_classes()`, `trend_line()`, `by_density()`, `delay()`
6. Report	Save the figures in `figures/` and show the figures and tables in this notebook.	the code cells below

Geographical scale and temporal scale

Geographical scale. The study covers all 342 Dutch municipalities, on the municipal borders of 1 January 2025; municipalities that merged between 2022 and 2025 are combined. The municipality is the right level for this question, because municipalities are responsible for placing public charging points.

Temporal scale. The data is annual and covers 2022 to 2025. Electric cars are counted on 1 January (2022–2025) and public charging points in December (2022–2025). Sub-questions 1 and 2 each describe one dataset over its full period of three years. Sub-question 3 compares both datasets at the same moments, which is possible from the end of 2022 to the end of 2024 (see Data preparation).

Data preparation

The EV data (CBS, 1 January 2022–2025), public charging points (RVO, December 2022–2025), population (CBS, 1 January 2025) and municipal borders (CBS, 2025) are loaded and merged with `load()` in src/plot_maps.py. For sub-question 3, `load()` in src/charging_vs_evs.py adds the matched years and population density (CBS, 2025). The code cell below runs both; the scripts can also be run on their own from the repository root (see the README).

The main cleaning steps are:

- **Name harmonisation.** The datasets spell municipality names differently (for example `AaenHunze`, `Utrecht (gemeente)` and `Den Haag`), so names are compared

without spaces and suffixes.

- **Municipal mergers.** Municipalities that merged after 1 January 2022 (Weesp; Brielle, Hellevoetsluis and Westvoorne; Heerhugowaard and Langedijk) are added up, so that all years use the 2025 borders.
- **Public charging points** are the sum of regular public, regular semi-public and fast (> 50 kW) charging points. An empty cell means zero charging points of that type, except for six municipalities without any data (Beek, Hengelo, Laren, Middelburg, Rijswijk and Stein), which are left out.
- **Small numbers.** Municipalities with fewer than 25 electric cars at the start of the period are left out of the growth analyses, because a growth percentage from such a small base is meaningless. On 1 January 2022 these are the Wadden Islands except Texel; at the end of 2022 Ameland, Schiermonnikoog and Vlieland.
- **Matching the dates.** Charging points are counted in December, electric cars on 1 January. To compare them at the same moment, sub-question 3 combines the charging points of December of one year with the electric cars of 1 January of the next year: December 2022 with 1 January 2023, and so on, up to December 2024 with 1 January 2025. Both then describe the end of the year. Sub-question 3 therefore covers the end of 2022 to the end of 2024. Sub-questions 1 and 2 each describe one dataset and use its full period.

In [1]:
```
import sys

from IPython.display import Image, display

sys.path.insert(0, "src")
import charging_vs_evs
import plot_maps
from plot_maps import FIG_DIR


def show(path, width=520):
    """Show a saved figure in the notebook."""
    display(Image(filename=path, width=width))


# One row per municipality: EVs (1 January 2022 and 2025), charge points (December 2022
# and 2025), inhabitants and borders. Used for sub-questions 1 and 2.
gdf = plot_maps.load()
# The same municipalities with matched dates (end of 2022-2024) and population density.
# `analysis` holds the municipalities used in sub-question 3.
matched, analysis = charging_vs_evs.load()
gdf[["statnaam", "inhabitants", "evs_first", "evs", "charge_points_first",
"charge_points"]].head()
```

```
342 municipalities, 333 in the analysis set
```

Out[1]:

	statnaam	inhabitants	evs_first	evs	charge_points_first	charge_points
0	Groningen	244441	1150	2916	1011.0	2053.0
1	Almere	229574	2161	6489	1208.0	2700.0
2	Stadskanaal	32038	95	321	20.0	115.0
3	Veendam	27687	112	369	31.0	89.0
4	Zeewolde	24291	286	826	115.0	315.0

Results

Sub-question 1: How has the number of electric passenger cars developed across Dutch municipalities between 2022 and 2025?

In [2]:
```
evs = plot_maps.ev_totals(gdf)
print(f"Electric cars: {evs[2022]:,.0f} on 1 January 2022, {evs[2025]:,.0f} on 1 January 2025 "
      f"({evs[2025] / evs[2022] - 1:+.0%})")
ev_maps = plot_maps.plot_ev_maps(gdf)
show(ev_maps[0], width=480)
```

```
Electric cars: 137,123 on 1 January 2022, 360,572 on 1 January 2025 (+163%)
Saved figure to figures/evs_per_1000_inhabitants_map.png
Saved figure to figures/evs_growth_map.png
```

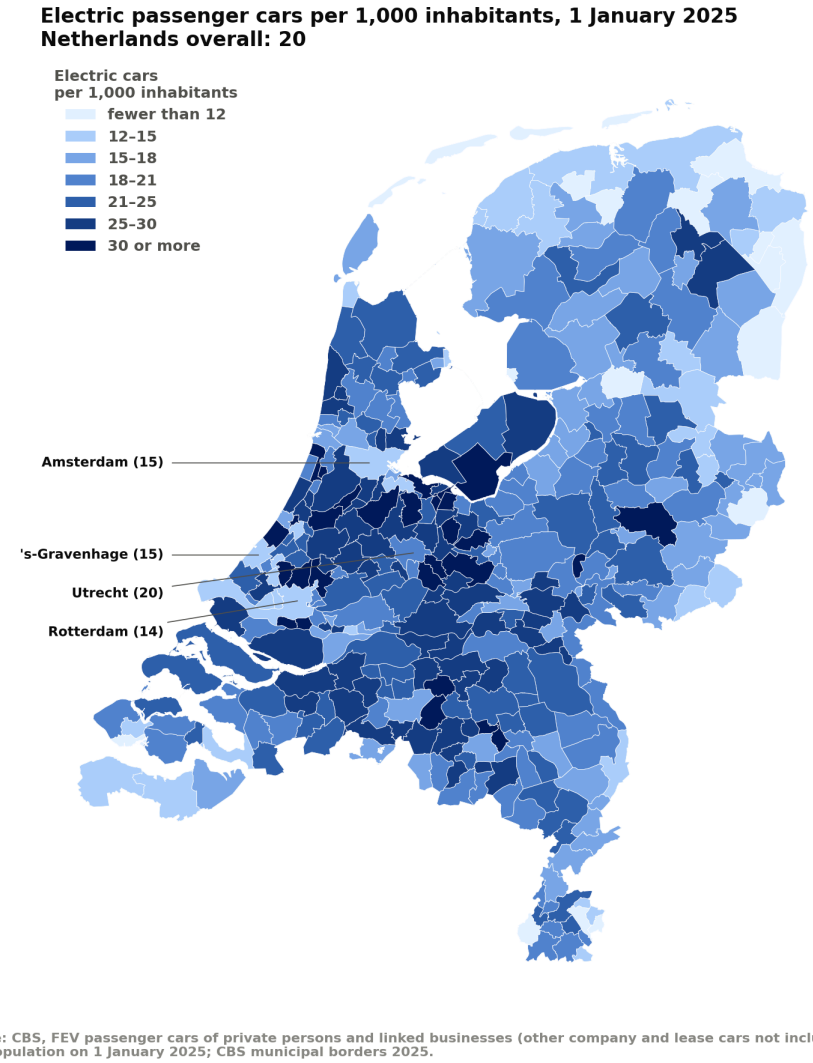


Figure 1. Electric passenger cars per 1,000 inhabitants per municipality, 1 January 2025.

In [3]:
```
show(ev_maps[1], width=480)
```

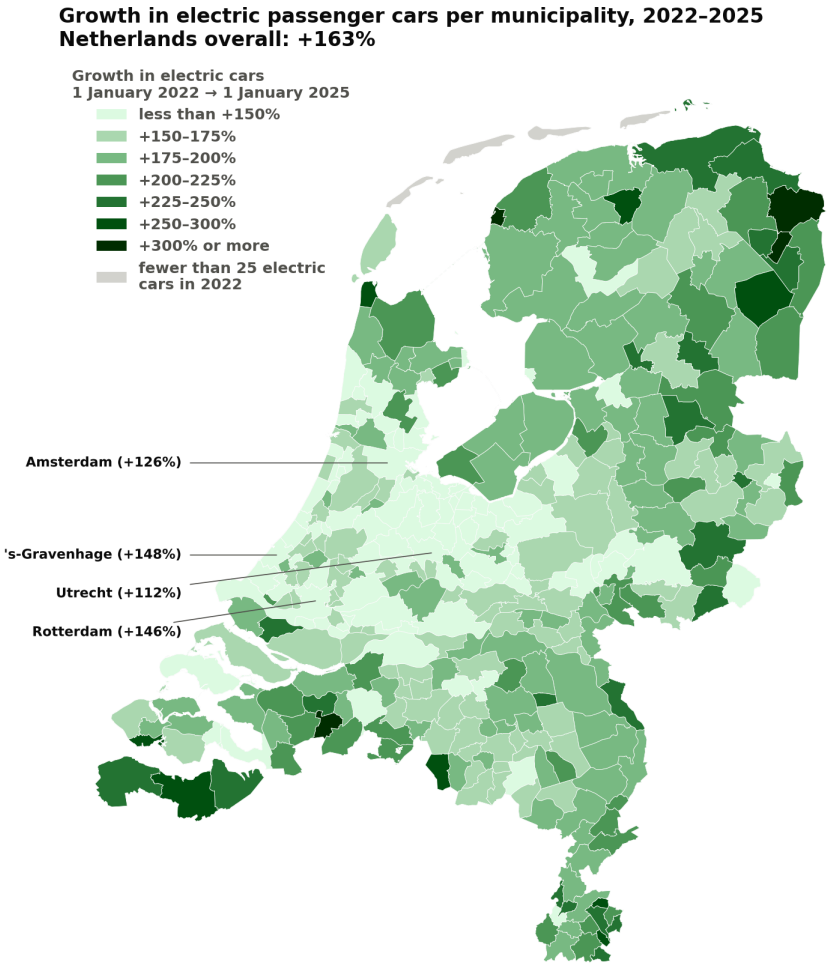


Figure 2. Growth in the number of electric passenger cars per municipality, 1 January 2022 – 1 January 2025.

Analysis

The number of electric passenger cars more than doubled in every municipality that had at least 25 electric cars in 2022. Nationally it grew from about 137,000 on 1 January 2022 to about 361,000 on 1 January 2025 (+163%), reaching 20 electric cars per 1,000 inhabitants.

EV ownership per inhabitant is highest in the suburban municipalities around Amsterdam, Utrecht and 's-Gravenhage, and lowest in the north-east and on the Wadden Islands (Figure 1).

The growth map (Figure 2) shows a different pattern. The Randstad grew relatively slowly (Utrecht +112%, Amsterdam +126%, Rotterdam +146%, 's-Gravenhage +148%), while the north-east, Zeeuws-Vlaanderen and parts of Limburg grew fastest, often by more than +225%. The municipalities with the most EVs per inhabitant are therefore not the ones where the fleet grew fastest: EV adoption is spreading from the early-adopting urban and suburban areas to the rest of the country.

Sub-question 2: How has the number of public charging points developed across Dutch municipalities between 2022 and 2025?

Public charging points are the sum of regular public, regular semi-public and fast (> 50 kW) charging points in the RVO data, counted in December of each year. Private charging points at homes and companies are not included.

In [4]:
```
plot_maps.charge_point_totals().style.format("{:,.0f}")
```

Out[4]:

December	Regulier Publiek	Regulier Semi-publiek	Snellaadpunten (>50 kW)	Total
2022	51,671	31,216	2,256	85,143
2023	69,829	40,386	3,007	113,222
2024	92,989	47,222	4,176	144,387
2025	108,481	72,883	5,733	187,097

Table 1. Public charging points in the Netherlands per type, December 2022–2025 (without the six municipalities that have no data).

In [5]:
```
charge_point_maps = plot_maps.plot_charge_point_maps(gdf)
show(charge_point_maps[0], width=480)
```

```
Saved figure to figures/charge_points_map.png
Saved figure to figures/charge_points_per_1000_inhabitants_map.png
Saved figure to figures/charge_points_growth_map.png
```

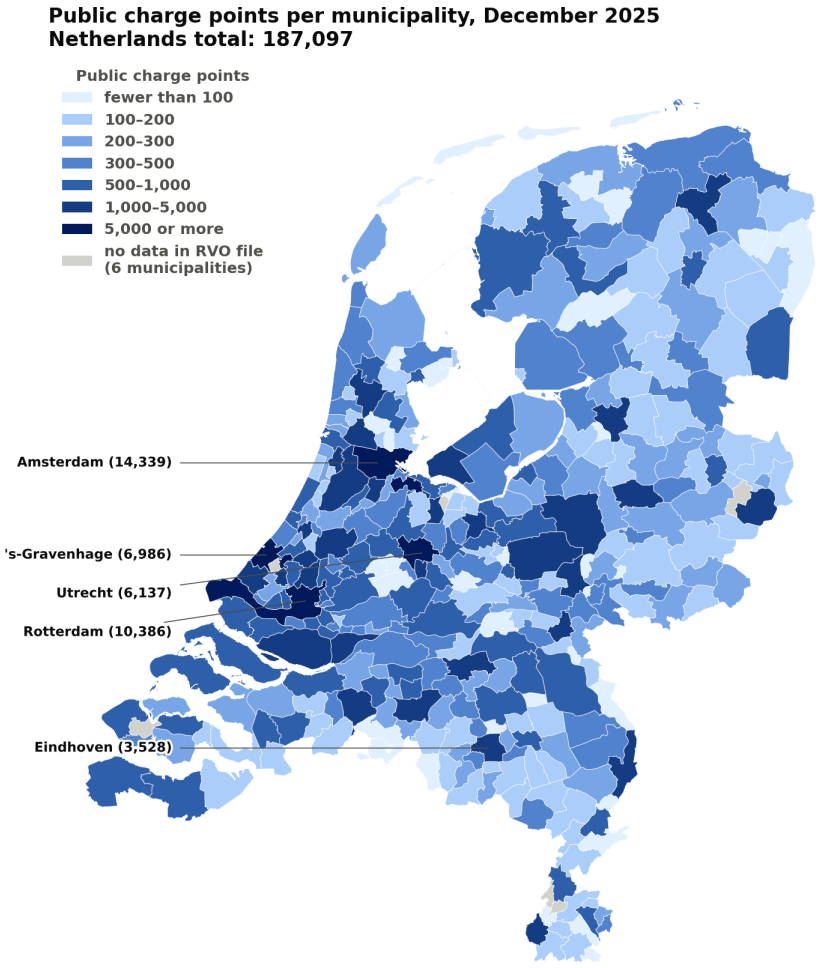


Figure 3. Public charging points per municipality, December 2025.

In [6]:
```
show(charge_point_maps[1], width=480)
```

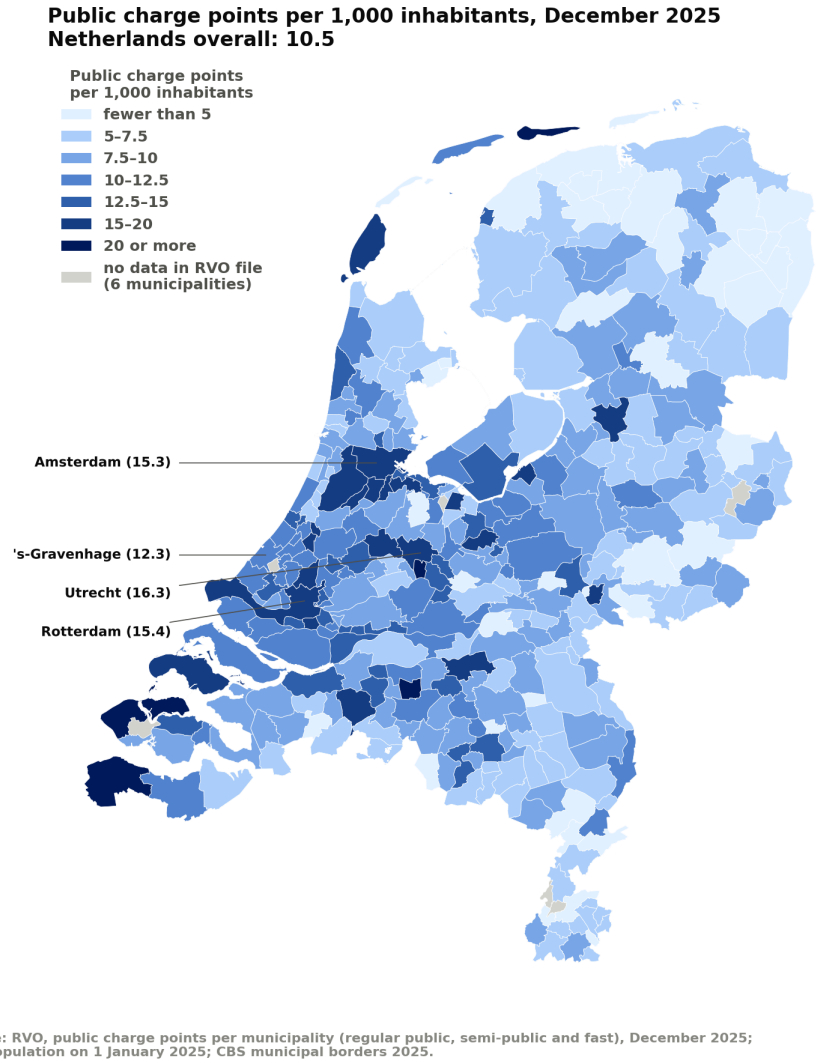


Figure 4. Public charging points per 1,000 inhabitants per municipality, December 2025.

```
In [7]: show(charge_point_maps[2], width=480)
```

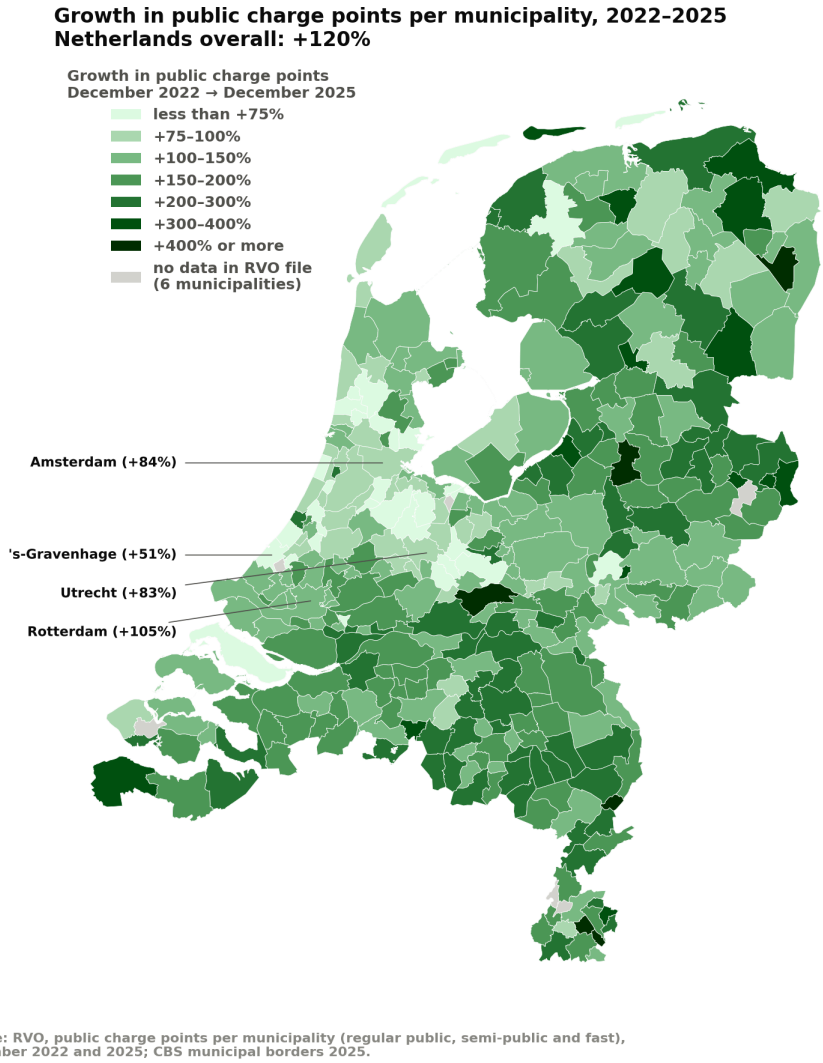


Figure 5. Growth in the number of public charging points per municipality, December 2022 – December 2025.

Analysis

The number of public charging points grew from about 85,000 in December 2022 to about 187,000 in December 2025 (+120%). Regular public charging points grew from 51,700 to 108,500, semi-public points from 31,200 to 72,900 and fast charging points from 2,300 to 5,700 (Table 1). These growth figures describe the charging points on their own and cannot be compared directly with the growth of electric cars in sub-question 1 (+163%): electric cars are counted on 1 January and charging points in December, almost a year later. Sub-question 3 compares both at the same moments.

In absolute numbers the charging points are concentrated in the four large cities (Figure 3: Amsterdam 14,300, Rotterdam 10,400, 's-Gravenhage 7,000, Utrecht 6,100). Per inhabitant the picture is different (Figure 4): the national average is 10.5 per 1,000 inhabitants, but the highest values are in tourist municipalities on the Zeeland coast and the Wadden Islands (Sluis 41.6, Noord-Beveland 29.0, Veere 22.8). These charging points mainly serve visitors,

not residents. The lowest values are in the north-east, for example Pekela (1.1) and Dantumadiel (1.4).

The growth map (Figure 5) mirrors the EV growth map (Figure 2): growth was slowest in the Randstad ('s-Gravenhage +51%, Utrecht +83%, Amsterdam +84%), which already had a dense network in 2022, and fastest in the more rural municipalities in the east and south.

Six municipalities (Beek, Hengelo, Laren, Middelburg, Rijswijk and Stein) have no data in the RVO file. They probably have names that exist in more than one province, so they are shown as missing rather than as zero. They are also missing from the national totals above.

Sub-question 3: Which Dutch municipalities show a match or mismatch between charging infrastructure availability and EV adoption between the end of 2022 and the end of 2024?

To compare charging points and electric cars at the same moment, this sub-question uses the charging points of December and the electric cars of 1 January of the following year (see Data preparation). It covers the end of 2022 to the end of 2024, for the 333 municipalities with charging-point data and at least 25 EVs at the end of 2022.

Four analyses compare the EV data with the charging-point data:

1. **EVs per public charging point (CP)** at the end of 2022 and 2024. If this ratio rises by more than a small margin (see below), charging infrastructure has not kept pace with EV growth.
2. **Trend line.** Do municipalities where the number of EVs grew more also add more charging points? Both increases are taken per 1,000 inhabitants, because in absolute numbers both mainly measure the size of a municipality. A trend line through all municipalities gives the expected number of new charging points for each municipality's EV growth. Municipalities far above or below the line are where charging and EV growth do not match.
3. **Population density.** In rural municipalities more people can charge at home on their own driveway, so fewer public charging points per EV are needed. Comparing municipalities by density checks whether a mismatch is real or just a difference in parking.
4. **Delay.** Charging points can follow EVs a year later. Section 3.4 checks whether the mismatches found are only such a delay.

What counts as a match or a mismatch. Even when charging infrastructure keeps pace, public charging points can follow electric cars with a delay. Under the Ladder van Laden, residents without their own parking space usually request a public charging point when they have an electric car, and placing it takes time. A small rise in the number of EVs per charging

point is therefore normal and not a mismatch in itself. This report uses two measures, each with a margin for this delay:

- **EVs per public charging point** (section 3.1). A change of less than 10% between the end of 2022 and the end of 2024 counts as a match. A rise of more than 10% means that charging fell behind the growth of EVs; a fall of more than 10% means that charging grew faster than the number of EVs (it caught up or went ahead). The margin is a choice: it is about three times the national rise per year (about 3%, Table 2), so small differences in when charging points are placed do not count as a mismatch.
- **New charging points compared with the trend line** (section 3.2). 80–125% of the number expected from EV growth counts as a match; less than half or more than twice that number is a clear mismatch.

A mismatch in either direction matters: too few charging points make electric driving harder for residents without a driveway, and too many use public money and space where they are less needed (see Introduction).

3.1 EVs per public charging point, end of 2022 – end of 2024

In [8]:
```
national, changed = charging_vs_evs.evs_per_charge_point(matched, analysis)
kept_pace = (analysis["evs_per_cp_change"] <= 0).sum()
print(f"Charge points kept pace with EVs (ratio did not rise) in {kept_pace} of "
      f"{len(analysis)} municipalities")
print(charging_vs_evs.match_classes(analysis).to_string())
national.style.format({"Electric cars": "{:,.0f}", "Public charge points": "
{:,.0f}",
                       "EVs per charge point": "{:.2f}"})
```

```
Saved figure to figures/evs_per_charge_point_map.png
Charge points kept pace with EVs (ratio did not rise) in 167 of 333 municipalities
Change in EVs per charge point, end of 2022 → end of 2024
went ahead (fell by more than 10%)     115
match (changed by less than 10%)       113
fell behind (rose by more than 10%)    105
```

Out[8]:

End of year	Electric cars	Public charge points	EVs per charge point
2022	198,193	85,143	2.33
2023	272,434	113,222	2.41
2024	356,126	144,387	2.47

Table 2. Electric cars, public charging points and electric cars per public charging point in the Netherlands at the end of 2022, 2023 and 2024 (municipalities with charging-point data).

In [9]:
```
changed.style.format({"EVs per CP end 2022": "{:.2f}", "EVs per CP end 2024": "
{:.2f}",
                      "Change": "{:+.0%}"})
```

Out[9]:

	Municipality	EVs per CP end 2022	EVs per CP end 2024	Change
Charging fell behind most	Terschelling	0.52	1.21	+131%
Charging fell behind most	Oldambt	3.15	5.84	+86%
Charging fell behind most	Uithoorn	2.58	4.28	+66%
Charging fell behind most	Zandvoort	2.49	4.03	+62%
Charging fell behind most	Vaals	2.13	3.45	+62%
Charging caught up most	Voerendaal	41.33	5.00	-88%
Charging caught up most	Simpelveld	14.14	4.85	-66%
Charging caught up most	Buren	15.44	5.57	-64%
Charging caught up most	Olst-Wijhe	14.13	6.09	-57%
Charging caught up most	Goirle	6.26	3.06	-51%

Table 3. The five municipalities where the number of electric cars per public charging point rose most (charging fell behind) and fell most (charging caught up), end of 2022 – end of 2024.

In [10]:
```
show(FIG_DIR / "evs_per_charge_point_map.png", width=900)
```

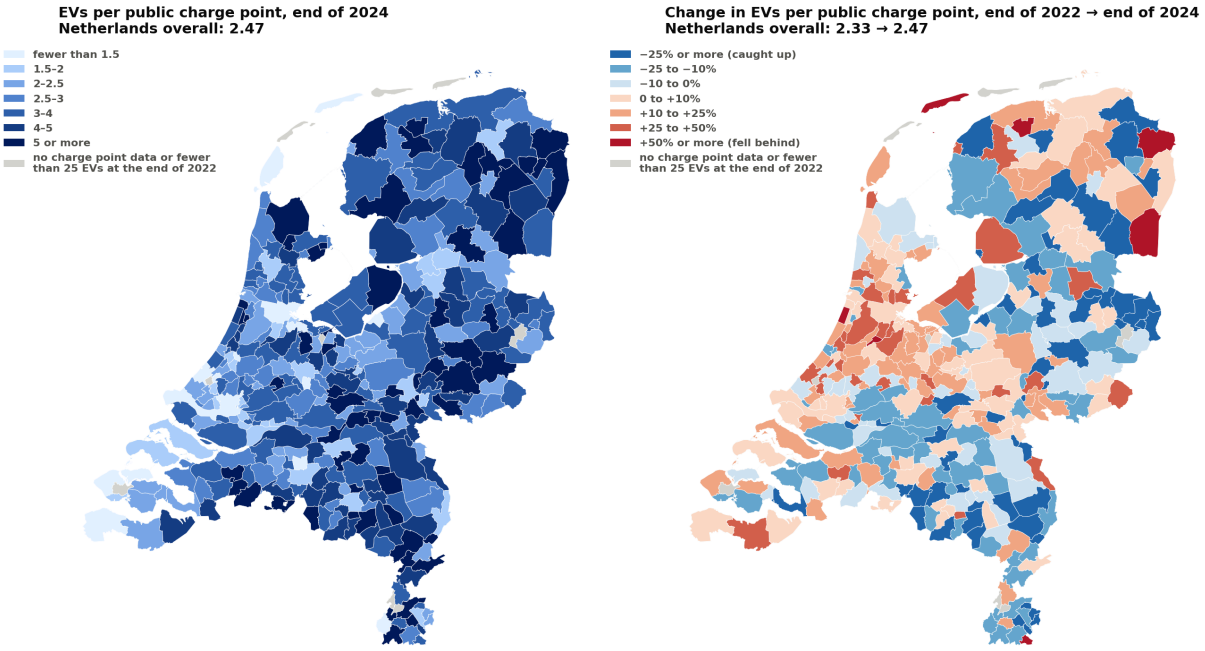


Figure 6. Electric cars per public charging point per municipality at the end of 2024 (left) and its change since the end of 2022 (right).

Analysis

Nationally, the number of electric cars per public charging point rose from 2.33 at the end of 2022 to 2.47 at the end of 2024, because the number of EVs grew faster (+80%) than the number of public charging points (+70%) (Table 2). The ratio rose in both years. Charging points grew at least as fast as EVs in 167 of the 333 municipalities; in the other 166 the ratio went up. With the 10% margin for delay, 113 municipalities show a match, 105 fell behind (the ratio rose by more than 10%) and 115 went ahead (it fell by more than 10%).

Figure 6 and Table 3 show both sides. The ratio rose most in Oldambt in the north-east of Groningen, in Uithoorn and Zandvoort in Noord-Holland and in Vaals in Limburg; in Oldambt there are now almost 6 EVs per public charging point. Terschelling is a special case: it has few electric cars (47) and lost charging points (from 48 to 39). The ratio fell most in municipalities that had very few charging points at the end of 2022, such as Voerendaal (41 to 5.0 EVs per charging point), Simpelveld, Buren and Olst-Wijhe. These municipalities caught up, while many municipalities that already had a dense network fell behind.

3.2 Trend line: does charging-point growth follow EV growth?

Each dot in Figure 7 is a municipality. The horizontal axis shows the increase in electric cars per 1,000 inhabitants between the end of 2022 and the end of 2024, the vertical axis the increase in public charging points per 1,000 inhabitants. Dividing by the number of inhabitants is necessary: in absolute numbers, large municipalities have both many new EVs and many new charging points simply because they have many residents. The line is the

trend through all municipalities (a linear regression). Figure 8 compares each municipality's new charging points with the number expected from the trend line.

In [11]:
```
fit, outliers, classes = charging_vs_evs.trend_line(matched, analysis)
print(f"R² per 1,000 inhabitants: {fit['R² (per 1,000 inhabitants)']:.2f}; "
      f"in absolute numbers: {fit['R² (absolute numbers)']:.2f}")
show(FIG_DIR / "charge_point_vs_ev_growth.png", width=640)
```

```
Saved figure to figures/charge_point_vs_ev_growth.png
Saved figure to figures/charge_point_growth_vs_trend_map.png
R² per 1,000 inhabitants: 0.01; in absolute numbers: 0.83
```

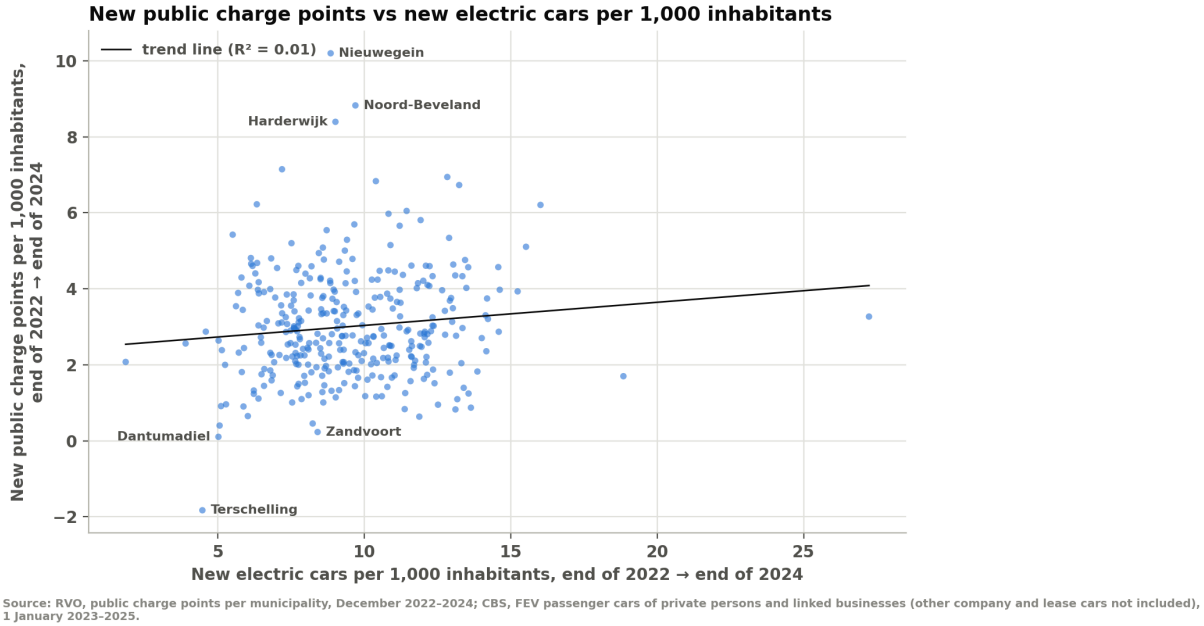


Figure 7. New public charging points against new electric cars per municipality, both per 1,000 inhabitants, end of 2022 – end of 2024, with the trend line.

In [12]:
```
show(FIG_DIR / "charge_point_growth_vs_trend_map.png", width=520)
```

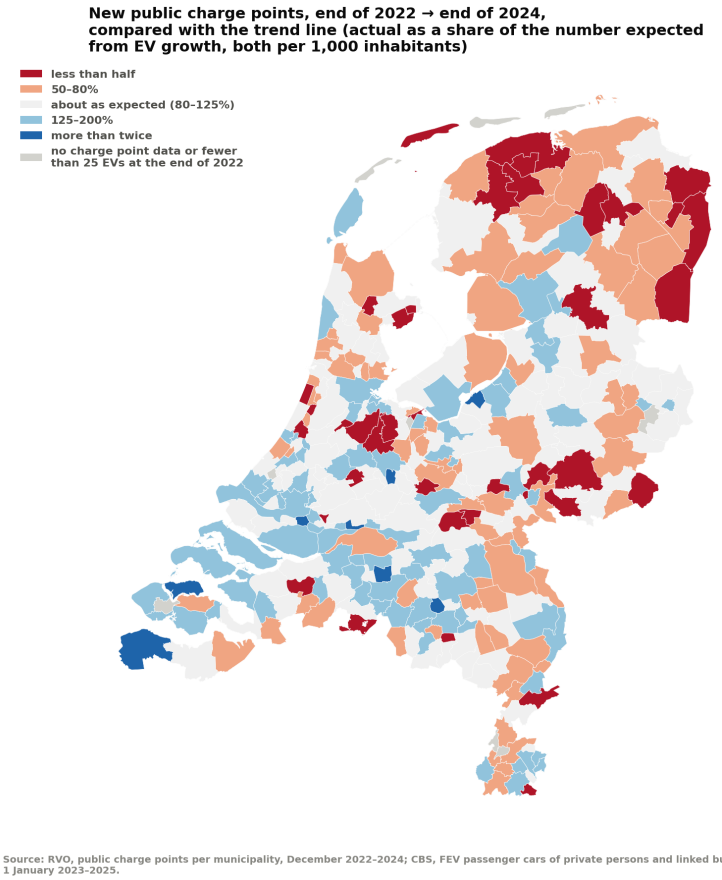


Figure 8. New public charging points per municipality, end of 2022 – end of 2024, as a share of the number expected from the trend line in Figure 7.

In [13]:
```
classes.to_frame()
```

Out[13]:

New charge points compared with the trend line	Municipalities
less than half	39
50–80%	80
about as expected (80–125%)	124
125–200%	82
more than twice	8

Table 4. Number of municipalities per class of Figure 8.

In [14]:
```
outliers.style.format({"New EVs per 1,000 inh.": "{:.1f}", "New CPs per 1,000
inh.": "{:.1f}",
                       "Actual / expected": "{:.0%}"})
```

Out[14]:

	Municipality	New EVs per 1,000 inh.	New CPs per 1,000 inh.	Actual / expected
Most new charge points	Nieuwegein	8.8	10.2	344%
Most new charge points	Noord-Beveland	9.7	8.8	292%
Most new charge points	Harderwijk	9.0	8.4	282%
Most new charge points	Sluis	7.2	7.1	249%
Most new charge points	Loon op Zand	10.4	6.8	223%
Fewest new charge points	Terschelling	4.5	-1.8	-68%
Fewest new charge points	Dantumadiel	5.0	0.1	4%
Fewest new charge points	Zandvoort	8.4	0.2	8%
Fewest new charge points	Pekela	5.1	0.4	15%
Fewest new charge points	Oldambt	8.2	0.5	16%

Table 5. The five municipalities with the most and the fewest new public charging points compared with the trend line.

Analysis

The trend line in Figure 7 is almost flat and the dots are widely scattered around it: municipalities where the number of EVs per inhabitant grew more did not add more public charging points per inhabitant. EV growth explains only 1% of the differences in charging-point growth between municipalities (R^2 = 0.01). Without the correction for size the two increases seem closely linked (R^2 = 0.83), but that only reflects that large municipalities have more of both.

Because the trend line is almost flat, the expected number of new charging points is about the same for every municipality (roughly 2.5 to 4 per 1,000 inhabitants). Figure 8 therefore mainly shows where charging points were added faster or slower than in a typical municipality. 124 of the 333 municipalities added about as many charging points as expected (80–125%, Table 4). The clearest deviations are (Table 5):

- **More than twice as many new charging points as expected** (8 municipalities), such as Nieuwegein, Noord-Beveland, Harderwijk, Sluis and Loon op Zand. Noord-Beveland,

Sluis and Loon op Zand (home of the Efteling theme park) are tourist municipalities, where charging points also serve visitors rather than residents.

- **Less than half as many as expected** (39 municipalities), for example Dantumadiel, Zandvoort, Pekela and Oldambt, which added almost no charging points, and Terschelling, which lost charging points. In Dantumadiel, Pekela and Oldambt, in the north, charging infrastructure has not kept up with EV growth.

Figure 8 also shows a regional pattern: many municipalities in the south-west (Zeeland, Zuid-Holland and Noord-Brabant) added more charging points per inhabitant than expected, while many municipalities in the north (Groningen, Friesland and Drenthe) added fewer.

3.3 The role of population density

The 333 municipalities are split into four equal groups, from the least dense to the most dense. Figure 9 shows for each group the median number of EVs per public charging point at the end of 2022, 2023 and 2024; the dashed line is the national ratio.

In [15]:
```
density = charging_vs_evs.by_density(matched, analysis)
density.style.format({"Inhabitants per km²": "{:,.0f}", "EV growth": "{:+.0%}",
                      "CP growth": "{:+.0%}", "EVs per CP end 2022": "{:.2f}",
                      "EVs per CP end 2023": "{:.2f}", "EVs per CP end 2024": "
{:.2f}"})
```

```
Saved figure to figures/evs_per_charge_point_by_density.png
```

Out[15]:

Density group	Municipalities	Inhabitants per km²	EV growth	CP growth	EVs per CP end 2022	EVs per CP end 2023	EVs per CP end 2024
least dense 25%	84	168	+84%	+100%	4.37	4.54	4.20
2nd quarter	83	359	+81%	+82%	3.98	4.11	3.57
3rd quarter	83	699	+80%	+80%	3.05	3.34	3.31
most dense 25%	83	2,070	+79%	+68%	2.23	2.37	2.44

Table 6. Medians per population-density group: growth of electric cars and public charging points, and electric cars per public charging point, end of 2022 – end of 2024.

In [16]:
```
show(FIG_DIR / "evs_per_charge_point_by_density.png", width=640)
```

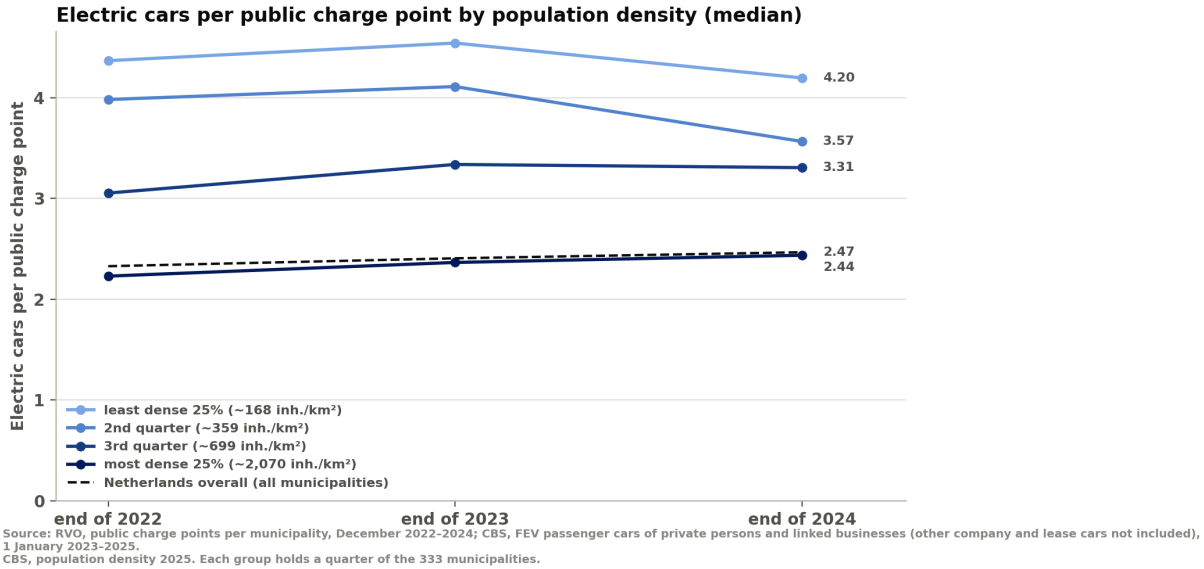


Figure 9. Median number of electric cars per public charging point per population-density group, end of 2022 – end of 2024.

Analysis

Sparsely populated municipalities have more EVs per public charging point (Figure 9): at the end of 2024 the median is 4.20 in the least dense group and 2.44 in the most dense group.

The change since the end of 2022 differs by group (Table 6). In the two least dense groups the ratio went down (4.37 to 4.20 and 3.98 to 3.57), because public charging points grew at least as fast as EVs. In the most dense group it rose every year, from 2.23 to 2.44: charging points there grew by a median of 68%, while EVs grew by 79%. EV growth was almost the same in all groups (+79% to +84%), so the difference comes from the growth of charging points.

This seems to contradict section 3.2, where many municipalities in the densely populated Zuid-Holland added more charging points than expected (Figure 8). Both are true. The most dense municipalities added the most charging points per inhabitant: a median of 3.4 per 1,000 inhabitants, against 2.5 in the least dense group. Because EV growth per inhabitant was about the same in both groups (a median of 8.6 new electric cars per 1,000 inhabitants), they also added more charging points per new electric car. But they started with a network twice as dense (a median of 5.0 charging points per 1,000 inhabitants at the end of 2022, against 2.5), so the same additions are a smaller percentage of what was already there. The least dense municipalities started with many EVs per charging point and the most dense with few, and the groups are growing towards each other.

In conclusion, the urban municipalities still have the most public charging points per EV. However, this is also where the number of EVs per charging point rose between 2022 and 2024. This matters because city residents usually cannot charge at home and depend on

public charging points. In rural municipalities the higher number of EVs per charging point is less of a problem, because more people can charge on their own driveway (RVO et al., 2026).

3.4 Is the mismatch only a delay?

Charging points can follow electric cars with a delay: residents usually request a public charging point only after they have an electric car, and placing it takes time. If the mismatches above were only such a delay, two things would be visible. First, the new charging points of a year would follow the new EVs of the year before, so a trend line with a delay of one year would fit better than one within the same year (Table 7). Second, municipalities that fell behind in the first year would catch up in the second year (Table 8).

In [17]:
```
delay_fits, catch_up = charging_vs_evs.delay(analysis)
delay_fits.style.format({"Slope": "{:.3f}", "R²": "{:.3f}"})
```

Out[17]:

	Slope	R^2
New charge points against new EVs (per 1,000 inhabitants)		
Same year, end of 2022 → end of 2023	0.005	0.000
Same year, end of 2023 → end of 2024	0.085	0.015
One year later: EVs end of 2022 → end of 2023, CPs end of 2023 → end of 2024	0.107	0.023

Table 7. Trend lines of new public charging points against new electric cars per municipality (both per 1,000 inhabitants), within the same year and with a delay of one year.

In [18]:
```
catch_up.to_frame()
```

Out[18]:

	Municipalities
Fell behind in the first year (ratio rose by more than 10%)	112
… of which the ratio went down again in the second year	69
… of which it went down by more than 10%	40
For comparison: all municipalities where the ratio went down in the second year	187

Table 8. Municipalities where the number of electric cars per public charging point rose by more than 10% from the end of 2022 to the end of 2023, and what happened in the following year.

Analysis

A delay explains only a small part of the mismatch. With a delay of one year, the trend line still explains only 2% of the differences in charging-point growth between municipalities (R^2

= 0.02, Table 7), hardly more than within the same year (R^2 = 0.00 and 0.02). Municipalities where the number of EVs grew more in one year did not add clearly more charging points in the next year either.

Some municipalities that fell behind did catch up (Table 8). Of the 112 municipalities where the ratio rose by more than 10% in the first year, it went down again in 69 (62%) in the second year, and by more than 10% in 40. But among all municipalities the ratio went down in 187 of 333 (56%) in the second year, so municipalities that fell behind caught up only slightly more often than others.

The national figures point the same way. If charging points simply followed EVs a year later, they would have grown in 2024 by about as much as the number of EVs grew in 2023 (+37%). They grew by 28% (Table 2), which is also less than the growth of EVs in the same year (+31%). The national rise in EVs per charging point is small (2.33 to 2.47), and part of it may be such a delay. The differences between municipalities are not: where charging points are added depends on other factors than local EV growth, in the same year or the year before.

Conclusion

This project examined to what extent public charging infrastructure in Dutch municipalities has kept pace with the growth of electric passenger cars between 2022 and 2025.

Sub-question 1: development of electric passenger cars. The number of electric passenger cars grew from about 137,000 on 1 January 2022 to about 361,000 on 1 January 2025 (+163%) and more than doubled in every municipality with at least 25 EVs in 2022. EV ownership per inhabitant is highest around the large cities, but growth was fastest in the more rural and peripheral regions, such as the north-east, Zeeuws-Vlaanderen and Limburg. EV adoption is spreading from the early-adopting urban areas to the rest of the country.

Sub-question 2: development of public charging points. The number of public charging points grew from about 85,000 in December 2022 to about 187,000 in December 2025 (+120%). Most charging points are in the large cities, but per inhabitant the highest values are in tourist municipalities on the coast and the lowest in the north-east. As with EVs, growth was slowest in the Randstad, which already had a dense network in 2022, and fastest in rural municipalities.

Sub-question 3: match and mismatch. Compared at the same moments (end of 2022 to end of 2024), the growth of charging points does not follow the growth of EVs at municipal level. After correcting for the number of inhabitants, municipalities where the number of EVs grew more did not add more public charging points (R^2 = 0.01). Where charging points were added therefore depends on other factors than local EV growth. As a result, the growth did not match in many places. With a margin of 10% for the normal delay between electric cars

and charging points, 113 of the 333 municipalities show a match, 105 fell behind and 115 went ahead:

- In tourist municipalities (Noord-Beveland, Sluis, Loon op Zand) charging points grew much faster than local EV numbers, because they also serve visitors.
- In some municipalities, especially in the north (Oldambt, Pekela, Dantumadiel) and parts of Noord-Holland (Uithoorn, Zandvoort), charging points fell clearly behind EV growth.
- Municipalities that had very few charging points at the end of 2022 caught up. In the dense urban municipalities the median number of EVs per charging point rose from 2.23 to 2.44 (+9%), just under the 10% margin, whereas it went down in the least dense municipalities.
- The mismatch is not only a delay: the new charging points of a year follow the EV growth of the year before hardly better than that of the same year (R^2 = 0.02), and municipalities that fell behind in 2023 caught up in 2024 only slightly more often than other municipalities.

Main research question. Public charging infrastructure has only partly kept pace with the growth of electric passenger cars. Charging points grew strongly, but not as fast as the number of EVs: between the end of 2022 and the end of 2024, EVs grew by 80% and public charging points by 70%. These matched figures are lower than those of sub-questions 1 and 2 (+163% and +120%), which describe each dataset on its own over three years and are measured almost a year apart, so those two cannot be compared with each other. The municipalities where most charging points were added are not the ones where the number of EVs grew most. As a result, the number of EVs per public charging point rose from 2.33 to 2.47 nationally, and in half of the municipalities (166 of 333), in 105 of them by more than 10%. The national rise is small, and part of it may be the normal delay between buying an electric car and getting a public charging point. The differences between municipalities, however, are not explained by such a delay (section 3.4). The development is most worrying in the urban areas: they still have the most charging points per EV, but this is where the median number of EVs per charging point rose most (+9%, close to the 10% margin). Residents in urban areas depend most on public charging because they often cannot charge at home.

Discussion

Limitations of the data

The results show that the development of public charging infrastructure differs considerably between Dutch municipalities. However, several limitations of the datasets need to be considered when interpreting these findings.

First, the CBS data do not include all electric passenger cars. Certain company and lease cars are excluded, even though their drivers may also use public charging infrastructure. Furthermore, the RVO dataset includes both public and semi-public charging points. Semi-public chargers may have restricted access or opening hours, meaning they are not always available to everyone. Private charging points are also excluded, although these can reduce the need for public charging infrastructure. This is particularly relevant when comparing urban and rural municipalities, where access to private parking and home charging may differ.

Another limitation is that charging-point data are missing for six municipalities, which were excluded from the analysis. Smaller municipalities can also show large percentage changes due to relatively small increases in EVs or charging points. To limit this effect, municipalities with fewer than 25 EVs at the beginning of the period were excluded from the percentage-growth comparisons.

Limitations of the methodology

The timing of the datasets affects the comparison between EV growth and charging infrastructure. EV registrations are recorded on 1 January, while charging points are measured in December. Therefore, the growth rates calculated in sub-questions 1 and 2 cannot be compared directly. For sub-question 3, the datasets were aligned by matching December charging-point data with EV registrations from the following January. As a result, this comparison covers the end of 2022 to the end of 2024 rather than the full study period.

This period of two years is short. Section 3.4 shows that the mismatches are not only a delay of one year, but a longer delay cannot be tested with this data: municipalities that fell behind may still catch up later. The 10% margin used to separate a match from a mismatch is also a choice. With a different margin the number of municipalities in each group changes, although the municipalities with the largest changes (Table 3) stay the same.

The EV-to-charging-point ratio is a useful indicator for comparing infrastructure provision between municipalities. However, a higher ratio does not necessarily indicate a shortage of charging infrastructure, because the number of registered EVs is not the same as the demand for public charging. The analysis does not account for charging power, actual charger utilisation, waiting times or access to private charging. Furthermore, the analysis is conducted at the municipal level, meaning that differences between neighbourhoods are not visible.

The regression results also require careful interpretation. In absolute numbers, EV growth and charging-point growth show a strong relationship ($R^2 = 0.83$). However, when growth is expressed per 1,000 inhabitants, the explanatory power decreases substantially ($R^2 = 0.01$). This suggests that differences in municipality size contribute considerably to the strong relationship observed in absolute numbers. The low explanatory power of the second model indicates that EV growth per inhabitant alone explains little of the variation in charging-point

growth. Other factors, such as municipal policies, tourism and existing charging infrastructure, may also play a role.

Population density was used to compare municipalities and provide an indication of potential differences in private charging availability. However, population density does not directly measure access to private parking or home charging. Therefore, these findings should be interpreted as associations rather than causal relationships.

Recommendations and further research

Future research could improve the analysis by including private charging availability, charging power and actual charger utilisation. This would provide a more complete understanding of whether charging infrastructure meets local demand. Analysing smaller geographical areas, such as neighbourhoods, could also reveal differences that are not visible at the municipal level. A longer time series would show whether the municipalities that fell behind between 2022 and 2024 catch up later.

Additionally, including factors such as household income, local charging policies and tourism could help explain why municipalities with similar EV growth show different developments in charging infrastructure. Comparing municipalities that install charging points proactively with those that respond to individual requests could also provide useful insights.

For municipalities, the results suggest that EV ownership and charging infrastructure should be monitored together. Particular attention could be given to densely populated municipalities, where the EV-to-charging-point ratio increased, and municipalities in the north where charging infrastructure grew relatively slowly. However, decisions about installing additional chargers should also consider local accessibility, charging demand and the availability of private charging options.

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Contribution Statement

Eline: I contributed to the development of the research proposal, including formulating the main research question and sub-questions. I also assisted with coding, searched for relevant information to help answer (sub)question and contributed to writing the introduction. I also reviewed the report as a whole to improve some writing and consistency mistakes.

Hendrik: Contributed to formulating the research objective, data processing, coding, plotting, writing the results and the conclusion.

Anne Jente: Contributed to finding the project subject, the data and writing the project proposal. For final report contributed to reformulating the RQ's after the feedback presentation, and rewriting/giving feedback on the text/method of the final report.

Matthijs: Together with the rest of the group, I contributed to the python code and contributed to answering the sub-questions and wrote the Discussion section of the final report. I created and set up the GitHub repository for the project and contributed to the group's collaboration by reviewing changes in a pull request before they were merged into the main branch.

Donna: Together with the rest of the group, I worked on the development of the main research question and sub-questions. Furthermore, I worked on the coding, contributed to answering the sub-questions and wrote a part of the conclusion of the report.